Meghapranay Raparla

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EDUCATION

Amrita Vishwa Vidyapeetham 2024 – 2028
B.Tech in Computer Science and Engineering · CGPA: 8.34 / 10 Coimbatore, India

- Coursework: Data Structures & Algorithms, Operating Systems, DBMS, OOP, Computer Networks
- 145+ LeetCode problems solved — graphs, dynamic programming, trees, recursion

EXPERIENCE

Machine Vision Intern — Nordic Tech Design May 2026 – Jun 2026
Industrial bolt quality-inspection systems using telecentric-camera machine vision Remote

- Built a sub-pixel dimensional metrology engine measuring thread pitch, major/minor diameter, and hole-to-edge distances; validated to <2 µm repeatability across 40 runs — on par with commercial inspection systems
- Developed a golden-comparison surface-defect detector with image alignment to eliminate lighting false alarms; achieved 100% recall and 100% precision on validation bolt, detecting one pit missed by the human operator
- Added automated cross-drilled hole counter from video of rotating bolt; deduplication logic correctly counted 4 holes despite each hole appearing multiple times per revolution

RESEARCH EXPERIENCE

Undergraduate Researcher — AI Security 2025 – Present
Amrita Vishwa Vidyapeetham · Advisor: Dr. Harini N.

- Hybrid C2PA + CNN-ensemble media authentication pipeline: **94.2% detection accuracy**, F1=0.93 on 10,000+ adversarial deepfake samples (StyleGAN2, ProGAN, Stable Diffusion); cryptographic bypass reduces avg. latency by ~49%
- Evaluated adversarial robustness under FGSM / PGD attacks; manuscript in preparation

Undergraduate Researcher — Agricultural CV 2025 – Present
Amrita Vishwa Vidyapeetham · Advisor: Dr. Amit Agarwal

- Fine-tuned YOLOv8 on 5,000+ annotated field images; **91.7% mAP@0.5** (precision 89.3%, recall 87.1%) for in-field strawberry disease detection

PROJECTS

IFDMS — Concurrent Focus Monitoring System Python · asyncio · WebSockets · Flask

- asyncio WebSocket server on a background thread manages multiple simultaneous clients; synchronous main loop runs 10+ ML modules with thread-safe shared state and producer-consumer IPC

MicroJava AST Interpreter Java · JavaFX · Compiler Design

- Full compiler pipeline: Lexer → Parser → AST → SVG visualizer → step-by-step interpreter with breakpoint debugger and symbol table / memory simulation

GeoDroneAI — Autonomous Drone Delivery Simulator Python · A* · Geospatial

- A* pathfinding with weighted heuristics and dynamic obstacle re-planning for multi-drone fleet; **34% faster** delivery times vs. naive routing in simulation

SmartStore ERP — Enterprise Backend Java · Spring Boot · JWT · MySQL

- RESTful ERP with JWT auth, RBAC, and layered Controller → Service → Repository architecture over Hibernate + MySQL; 7 normalized tables, 7 REST endpoints

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, C
Concurrent Systems: asyncio, WebSockets, multi-threading, thread-safe shared state, producer-consumer IPC
AI / ML / CV: PyTorch, OpenCV, EfficientNet, YOLOv8, CNNs, ensemble learning, sub-pixel metrology, anomaly detection
Systems & Backend: Spring Boot, Flask, REST APIs, JWT/RBAC, MySQL, SQLite
Tools & Methods: Git, Linux, IntelliJ, VS Code, Agile/Scrum, DSA, Compiler Design